INCH-POUND
MIL-DTL-24643/34E
22 August 2002
SUPERSEDING
MIL-C-24643/34D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS TYPE LSMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, AWG Size No. 16, class B stranding, per ASTM B8, (7 by 0.0192 inch) or in accordance with ASTM B 286, AWG size No. 16 with19 strands, tin coated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation, white. Minimum average wall thickness 0.013 inch. Standard identification code applied by method 1.
Fourth -	The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter. Cabling sequence shall be consecutive starting with No. 1 from center outward.
Fifth -	Binder tape, applied helically with overlap.
Sixth -	Braided shield of tin coated copper.
Seventh -	Binder tape applied helically with overlap.
Eighth -	Cross-linked polyolefin jacket. Minimum average cable jacket thickness 0.075 inch.

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TABLE I. Details.

Military part no. 1/ M24643/34	Size	Number of conductors	Diameter over shielded core (inch nom.)	Overall diameter (inches) TYPE LSMS	
				min	max
-01UO	37	37	0.625	0.740	0.800

1/When double shield is required, see MIL-DTL-24643 for configuration of part number.

NOTE: Weight of completed cable shall not exceed 0.060 pound/foot.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 5.1
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
- Insulator resistance – megohms-1000 feet minimum
  - Conductor to conductor 2000
  - Conductor to shield 500
- Jacket flaws ..... No failure
- Conductor and shield continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
- Jacket (When tested at 200°C) 50
  - Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Insulation
  - Ethylene propylene
  - Air oven
    - Tensile strength - percent of unaged, minimum 75
    - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (conductor) - cycles minimum 25
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... No failure
- Surface transfer impedance Milliohms per meter, maximum 700
- EMP response time - dB, minimum 60

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
- Aging and compatibility (cable)(125 ± 5°C) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
- Halogen content - percent, maximum
  - Jacket 0.2
  - Fillers 0.2

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Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-032)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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